Predicting Baseflow with Linear Regression

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Project: https://github.com/ewalburger/CS5830_Project6
Presentation: cs5830-project6-presentation

Introduction

In this project we attempt to provide a model that can reasonably predict Baseflow using linear regression on the given Hydrological dataset. The dataset is from the Republican River Basin, where data was collected and analyzed to assess the impact of ground-water pumping for irrigation. Baseflow represents how much ground-water flows into nearby streams and rivers. It's a very important metric for looking at how reliable a source of water is. The more irrigation pumping is done, the less baseflow there is. Baseflow is also affected by factors that affect how much water is available such as precipitation, and evapotranspiration. Creating effective predictive models for baseflow is very important because it informs communities, farmers, and people in power of water usage patterns, and can predict how much water we will have in the future.

Dataset

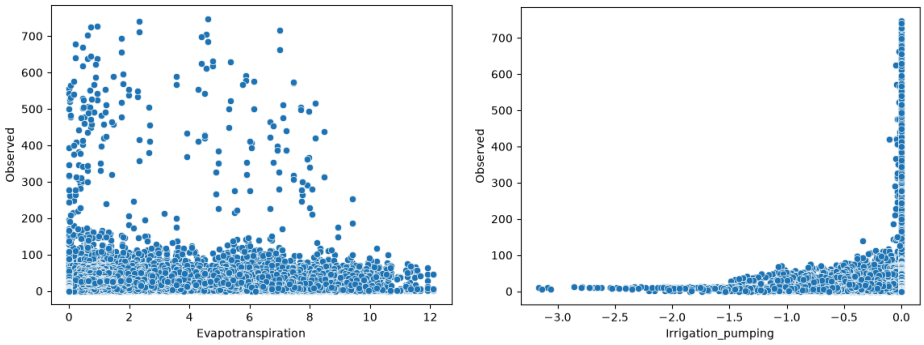
The dataset used contains measurements/observations of various hydrological attributes at different segments of a river on multiple different dates. It holds hydrological data from the "Republican River Basin [which covers] portions of eastern Colorado, northwest Kansas and southwest Nebraska" (ref 1). The dataset holds measurements and observations of this Republican River Basin area from 1939 up to 2000, increasing our confidence in the dataset as it spans such 61 years. The dataset did require some cleaning after some background research, we had to get the dates of measurements into a more readable format and remove some of the Baseflow observations which were not very useful (0 and negative).

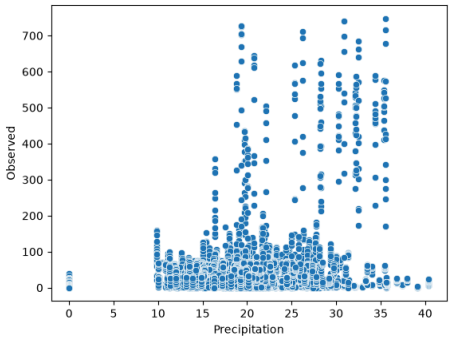
Analysis Technique

We used multiple linear regression on the dataset as we mainly compared quantitative variables, such as precipitation, irrigation pumping, and evapotranspiration against our target variable Baseflow, also quantitative. We needed to use multiple linear regression as baseflow appears to be affected by more than just one explanatory variable.

Results

Through the process of analysis, we were able to create a linear regression model using several different variables that had significant predictive power. However, we found linear regression by itself to be very limited in actually being able to predict baseflow. As we worked to find a good model, we analyzed different scatterplots to see relationships between variables The first scatterplot, see below, was of the relationship between evapotranspiration and observed baseflow. Visually, we noticed a small trend where as evapotranspiration goes up, baseflow goes down. In the dataset, the more negative irrigation pumping is, the more water is being removed from the river. The less irrigation pumping is occurring, the more baseflow there is, which makes sense because there is more water left over to flow into nearby streams and rivers. We also noticed that the more precipitation there is, the more baseflow there tends to be.





Understanding these relationships enabled us to create a predictive model for Baseflow using several of the variables. The predictive variables we found as the best predictors were evapotranspiration, precipitation, and irrigation pumping. We found that log-transforming baseflow led to a better predictive model. This means that the coefficients represent approximate percentage changes in baseflow, and we can interpret the coefficients using the percentage formula. The precipitation coefficient was .0748, which means that for every 1 unit increase in precipitation, the model predicts that baseflow will increase by about 7.5%. The p-value shows that this is highly significant, which makes sense logically as rain adds to there being more water in a system. The irrigation pumping coefficient was .5196, which takes some interpretation because the pumping data is stored as a negative number, where the more negative it is, the more pumping is happening. This coefficient shows that as irrigation pumping increases in intensity by 1 unit, baseflow decreases by 40.5%. This coefficient was also highly

significant. Evapotranspiration had a coefficient of .0228 which shows a slight positive relationship with 2.3% increase in baseflow per unit of evapotranspiration. This slight positive relationship is because Evapotranspiration peaks in the summer, so this is capturing a seasonal cycle pattern, rather than acting as a direct physical drain on the same day.

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                            OLS Regression Results
==============================================================================
Dep. Variable:       np.log(Observed)   R-squared:                       0.237
Model:                           OLS   Adj. R-squared:                  0.237
Method:                Least Squares   F-statistic:                     1413.
Date:               Thu, 29 Oct 2026   Prob (F-statistic):               0.00
Time:                       22:09:21   Log-Likelihood:                -23847.
No. Observations:              13637   AIC:                         4.770e+04
Df Residuals:                  13633   BIC:                         4.773e+04
Df Model:                          3
Covariance Type:            nonrobust
==============================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept              0.9794      0.023     43.248      0.000       0.935       1.024
Evapotranspiration     0.0228      0.005      4.929      0.000       0.014       0.032
Precipitation          0.0748      0.001     58.613      0.000       0.072       0.077
Irrigation_pumping     0.5196      0.050     10.317      0.000       0.421       0.618
==============================================================================
Omnibus:                      285.212   Durbin-Watson:                   0.104
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              304.515
Skew:                          -0.352   Prob(JB):                     7.51e-67
Kurtosis:                       3.204   Cond. No.                         81.4
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
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Overall, the intercept for this model is 0.9794 which is the baseline predicted baseflow when precipitation, ET, and pumping are all zero. The R squared value, which shows the model fit and explanatory power is 0.237 which means that this model explains 23.7% of the variance in the log-transformed baseflow. The model is not perfect, but does hold some explanatory power.

Technical

This was a very messy dataset, and it took a lot of trial and error to determine what formatting and cleaning needed to be done for analysis. We formatted the dates using the method recommended in the project description. We chose to remove all non-positive baseflow conditions. We originally got the idea from the paper that was provided “Data-driven methods to improve baseflow prediction of a regional groundwater model”. We researched groundwater and baseflow measurements more to see why this would be better for modeling. We found that negative baseflow can’t physically exist. Stream gauges and sensors have a margin of error, so when they’re dry you can get a slightly negative reading that can negatively impact regression models. Overall, it is also much more useful to analyze the water when it is actually flowing, because that informs us better of how irrigation pumping affects the baseflow. Another method we used for cleaning the data was that we only included gauging stations that have at least 10

years of monthly baseflow observations in order to better inform the model from gauging stations that have historical data.

There are many linear relationships when it comes to predicting baseflow, such as precipitation, evapotranspiration, and the amount of irrigation pumping. This makes linear regression suitable for this kind of data. However, reading the provided article also showed that there are limits to how predictive a linear regression model can truly be because a lot of the factors are not linear, such as time of the year. It is also hard to predict with a simple model how things like rainfall will impact baseflow, because those reactions typically take time.

The process we went through to find our results took a lot of trial and error. We looked at many scatterplots to determine relations between variables. We analyzed our data before and after it was cleaned to see if it really made a difference. We looked at a few different models with different combinations of explanatory variables. We ended up comparing their summaries to see which ones were best. We decided to try a log transformation of the baseflow, since baseflow is skewed, and that ended up being much better than baseflow by itself. We also compared the overall distributions between each column to make sense of some of the data, such as the relationship between irrigation pumping and baseflow. There were other things that we could have tried with more time and experience with this kind of analysis.Time series analysis could have been implemented to look at different factors over time instead of independently. This was not a technique we were familiar with, and with our limited time for the project, we didn't have time to look into it. But for other similar models, it appears to be very useful. It also felt like we could have done more to clean the data, but we couldn't determine any better strategies with the time we had.

References

1. Data-driven methods to improve baseflow prediction of a regional groundwater model by Tianfang Xu, Albert J. Valocchi